36 (SEM–II) AS 121

2026

CHEMICAL SCIENCE–II

Paper : AS–121

Full Marks : 100

Time : Three hours

The figures in the margin indicate full marks for the questions.

*Answer **all** questions.*

PART–A

1. Multiple Choice Questions : $1 \times 5 = 5$

(a) A battery is an example of—

(A) corrosion

(B) entropy

(C) energy conversion device

(D) metallurgy

(b) A chemical reaction shows the tendency to—

(A) attain high energy state

(B) rapid decay

(C) attain low energy state *

(D) obtain energy

(c) A device where chemical change is caused by electrical energy from external source is—

(A) Voltaic cell

(B) Potentiometer

(C) Voltameter

(D) Electrolytic cell *

(d) Gibb's free energy (G) is used to predict—

(A) spontaneity of a process *

(B) physical properties of a process

(C) polarity

(D) potential energy surfaces

(e) Charging a battery is—

(A) Spontaneous process

(B) Voltaic cell

(C) Endothermic process

(D) Non-spontaneous process *

2. Answer the following : ***(any three)*** $2 \times 3 = 6$

(a) What information does thermodynamic provide ?

(b) Why do metallic products corrode ?

(c) What do you understand by electromotive (emf) of a cell ?

(d) What are half cells ?

3. Answer the following : ***(any five)*** $5 \times 5 = 25$

(a) Explain the electrochemical nature of metallic corrosion.

(b) Explain how Gibb's free energy is related to metallurgy.

(c) Illustrate how Gibb's free energy can predict the acid-base equilibria ?

(d) Describe the function of an electrochemical cell.

(e) How do you predict feasibility of reaction with the help of the electrochemical series ?

(f) Show the relation between the maximum work derived in a chemical reaction and Gibb's free energy.

PART–B

4. Select the incorrect statement from the following options : 1

   (A) Racemic modifications are an equimolar mixture of dextrorotatory and levorotatory isomers

   (B) Racemic mixture is designated as dl-pair

   (C) Meso compounds are optically active

   (D) Meso compounds contain more than one chiral carbon centre

5. Optical isomers that are non-superimposable mirror images are known as— 1

   (A) Tautomers

   (B) Diastereomers

   (C) Enantiomers

   (D) Metamers

6. Using Slater's rules, calculate Z_{eff} for the following electrons : 3×2=6

   (i) an outermost electron of Cl

   (ii) a 3d electron in Mn

7. State Fajan's rules. Arrange the following species into increasing order of covalent character. 3+2=5

KCl, $CsCl$, $LiCl$, $NaCl$

8. Justify the following statements : ***(any two)*** 3×2=6

(i) $BeCl_2$ covalent while $BaCl_2$ is ionic.

(ii) AgI more stable than AgF.

(iii) Fe^{3+} prefer O-donor ligands over S-donor ligands.

9. State HSAB (Hard-Soft Acid-Base) principle. Identify the following species into their hard/soft acid/base category : 1+4=5

Pd^{2+}, Al^{3+}, F^-, I^-, H^+, H^-, Ag^+, Li^+, ^-OH,

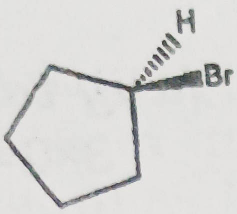
CO, CN^-, Cu^+

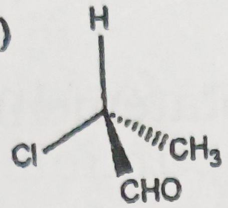
10. What are the uses of analgesic and antipyretic drugs? Write the chemical equation involved in the synthesis of paracetamol from phenol. 2+2=4

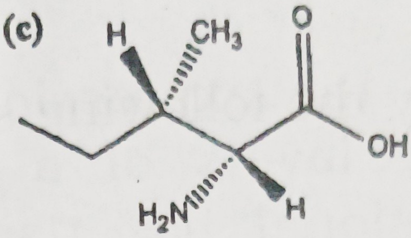
11. What do you mean by chirality of a molecule? Define diastereoisomers and enantiomers with example. 2+4=6

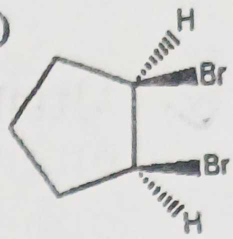
OR

Assign R/S configuration to the following molecules : $2\times3=6$

(a) 

(b) 

(c) 

(d) 

PART–C

12. Using Hückel's rule, explain whether the *cyclopentadienyl cation* is aromatic, antiaromatic, or non-aromatic. 3

13. Define nodal plane and discuss its significance in atomic orbitals. Draw the pictorial representation of the nodal points and nodal planes in 3*s* and 3*p* orbitals. 2+3=5

14. Explain the bonding in H_3^+ ion using molecular orbital theory and predict its structure with a suitable diagram. 5

15. Draw and explain the π-molecular orbital diagram of 1,3-butadiene, indicating the energy levels and nodal planes. 5

16. Compare the molecular orbital diagrams of CO and NO with respect to electronic configuration, bond order, and magnetic properties. 5

17. Derive the expression for the quantized energy levels of a particle in a one-dimensional box using the Schrödinger equation. Also predict the total energy for third energy level (i.e. $n = 3$). 5+2=7

$\nabla^2\psi + \frac{2\pi^2 m}{h}\frac{\partial\psi}{\partial t}$ 0

$i\frac{h}{2\pi}\frac{\partial\psi}{\partial x} = E\psi$

$\frac{m}{z}$ $\frac{m^2}{z^2}$ $\frac{z^2}{m^2}$